Model Performance Across Species

Four models, three evaluation regimes, from in-distribution discrimination to genome-scale deployment

August 2026 (0.6-clustering retrain)

Abstract

This report consolidates results for the four promoter-prediction models we now train (ProkBERT-mini, Nucleotide Transformer (NT) with LoRA, a from-scratch one-hot CNN, and CNN-v2, a multi-encoding fusion CNN that adds ENAC positional and di/tri-nucleotide features), broken down by bacterial species and across three evaluation regimes. All numbers here are from the stricter 0.6-identity clustering retrain (MMseqs2), so val/test splits are cluster-disjoint at 0.6. Each regime answers a different question and is reported separately, because they are not directly comparable. Regime A (in-distribution, heterogeneous negatives) measures ordinary discrimination on training-style data. Regime B (composition-matched negatives) forces the models to read the motif rather than base composition. Regime C (genome-scale precision) slides each model across a whole held-out genome to measure what deployment actually looks like: of the sites it flags, how many are real. Three findings stand out. First, CNN-v2 is the standout of this round: it has the highest composition-matched (SRS) AUROC of any model, which is the cleanest evidence of genuine motif learning rather than an AT-content shortcut, and it is the most precise at genome scale at mid-to-high recall (0.5 to 0.9) in the restricted scan, that is, it yields the fewest false positives per real promoter recovered. Second, discrimination on the curated test sets does not predict deployment behaviour, and the point is now visible twice over: NT places second on balanced test AUPRC (0.854) yet is the weakest model at genome scale, and ProkBERT is first on balanced AUPRC (0.884) yet falls to third or fourth in the restricted genome-scale scan, its high-confidence shortlist the least precise of the four. Third, the 0.6 retrain, which barely moved any model's balanced AUPRC, cost ProkBERT genome-scale precision specifically: its restricted precision at 50% recall dropped from 45% on the earlier 0.9-clustered checkpoint (T18, validation AUPRC 0.876) to 33% here, while NT and the CNN, run through the identical pipeline, held steady. Re-scanning that older checkpoint through the same pipeline reproduced its 45%, confirming the regression is real; because the two checkpoints differ in both the clustering threshold and the swept hyperparameters, it is attributable to the retrain as a whole rather than the threshold alone. The best standalone deployment configuration in this round is CNN-v2 with restrict-scan; the intended deliverable remains the planned four-model rank-percentile ensemble, and the deliverable it implies is an enriched candidate list, not a map.

1 Purpose

Aggregate metrics obscure where a model's performance comes from, and curated-set metrics obscure what happens at genome scale. This report therefore stratifies every result two ways: by species, to separate cross-species from in-genus performance, and by evaluation regime, to separate curated-set discrimination from genome-scale deployment. Because the target, *C. butyricum*, has no ground-truth promoters of its own, its in-genus relatives (*C. acetobutylicum*, *C. beijerinckii*, *C. ljungdahlii*) are the best available proxies for how the models will behave when mined on the target.

2 The three evaluation regimes (read this first)

The same model can look very different depending on what it is scored against and at what base rate, so keeping the three regimes distinct is essential to reading the tables correctly. They are ordered below by increasing difficulty and realism.

- **Regime A: in-distribution, heterogeneous negatives (Table 1).** From the T15 held-out test set, each species' verified positives are ranked against the shared pool of training-style negatives

(CDS, intergenic, downstream, and dinucleotide-shuffle windows from 166 *Clostridium* genomes). This measures ordinary discrimination on data drawn like the training data, and uses the easiest negatives. Metric: AUROC.

- **Regime B: composition-matched negatives (Table 2).** Each test positive is scored against a dinucleotide shuffle (identical base and dinucleotide composition, motif destroyed). Because the negative is composition-identical to the positive, base composition gives the model no help, so it must read the actual motif. We report the well-powered in-distribution form here (967 test positives against 611 SRS shuffles, all four models, 0.6 retrain). A separate, stricter out-of-genus generalization test (held-out T9 engineered promoters versus their shuffles, in-genus *C. acetobutylicum* against out-of-genus *C. phytofermentans*) is discussed at the end of the section; those numbers are now refreshed on the 0.6 models for all four. Metric: AUROC with 90% bootstrap CI; chance is 0.5.
- **Regime C: genome-scale precision (Table 3).** From a whole held-out in-genus genome (*C. ljungdahlii*), each model is slid across every window of the genome, where real promoters are only about 0.01% of positions. This is the only regime that shows the deployment reality, namely how pure a real scan's output is (what fraction of the sites it flags are real promoters), which the curated regimes structurally cannot. Metric: precision at fixed recall (equivalently, false-positive peaks per real promoter recovered), plus enrichment over random.

A model can be excellent in A, good in B, and still produce thousands of false positives in C. That is not a contradiction; the three regimes simply answer different questions. In particular, a model that wins in A and B need not win in C, because C is limited by the genome base rate, not by curated-set discrimination, and this round shows exactly that reordering.

3 Regime A: in-distribution performance by species

Table 1: AUROC on the T15 test set, by species, against the shared heterogeneous negative pool. (n = verified positives in the test split; **bold** = best model per species.)

Species (n_{pos})	ProkBERT	NT-LoRA	CNN	CNN-v2
E. coli (404)	**0.933**	0.929	0.878	0.920
C. beijerinckii (360)	**0.947**	0.924	0.914	0.936
C. acetobutylicum (166)	**0.946**	0.921	0.908	0.912
B. subtilis (37)	0.897	**0.936**	0.885	0.884

All four models are strong across every species (AUROC 0.88–0.95), so this regime mainly establishes a floor. Two patterns matter for the target. First, ProkBERT's two best species are the in-genus *Clostridium* (0.947 and 0.946), so its in-distribution performance is not carried by the cross-species data, which is a good sign for transfer to *C. butyricum*. Second, CNN-v2 is clearly the stronger of the two CNNs here, matching the transformers on the in-genus species (0.936 on *C. beijerinckii*, second only to ProkBERT) and improving on the from-scratch CNN on most species, by up to four points on *E. coli*. NT-LoRA's single best species is again the cross-species *B. subtilis* (0.936), slightly above its in-genus scores, so its strength is not a uniform cross-species advantage. Recall at the 0.5 operating point is highest for CNN-v2 on three of the four species (0.84–0.94, the exception being *C. acetobutylicum*, where ProkBERT is higher), but that reflects CNN-v2's systematically higher scores (its raw output is optimistic, see Regime B and the negatives-by-class breakdown) rather than better discrimination; recall at a fixed threshold is not a discrimination metric and should be read alongside AUROC, not as confirmation. Because these are the easiest negatives, strong numbers here are necessary but not sufficient, and Regimes B and C do the real discriminating.

4 Regime B: composition-matched (motif-only) discrimination

Because SRS negatives are composition-matched, separability here can only come from the promoter motif, not from base or GC composition, so this is the cleanest motif-learning test in the balanced setting.

Table 2: Test positives (967) against their dinucleotide-shuffle negatives (611 SRS windows; identical base and dinucleotide composition, motif destroyed). AUROC with 90% bootstrap CI. Chance = 0.5. **Bold** = best.

Model	AUROC (positives vs SRS shuffle) [90% CI]
ProkBERT	0.906 [0.90–0.91]
NT-LoRA	0.896 [0.89–0.91]
CNN	0.896 [0.89–0.91]
CNN-v2	**0.924 [0.92–0.93]**

All four models clear chance decisively (0.90–0.92), which is reassuring for a project whose central risk is AT-content false positives: none of the four is separating positives from shuffles on composition alone. CNN-v2 is the best, at 0.924, above ProkBERT (0.906) and clearly above NT and the from-scratch CNN (both 0.896), with a CI that does not overlap the lower three. The ENAC positional features appear to give CNN-v2 a genuine read on motif structure, and this is the single most encouraging number in the report: the model that is only third on balanced AUPRC is first at the one test designed to isolate motif signal. Across the other three negative classes (CDS, intergenic, downstream; full breakdown in `negatives_by_class.csv`), CDS is the easiest for every model (AUROC 0.925–0.980) and the structured genomic negatives are all more separable than SRS, as expected.

Out-of-genus generalization (0.6 models, four-way). A stricter test uses the held-out T9 engineered promoters scored against their own shuffles, split into in-genus *C. acetobutylicum* (n=68) and out-of-genus *C. phytofermentans* (n=25, reclassified into *Lachnoclostridium*). In-genus, ProkBERT is clearest of chance (AUROC 0.736 [0.65–0.80]), with both CNNs also above it (CNN-v2 0.659 [0.57–0.74], CNN 0.629 [0.54–0.71]); NT-LoRA, by contrast, sits at chance (0.440 [0.36–0.53]), a marked drop from its best-in-class 0.71 on the pre-0.6 models and consistent with its weakness elsewhere this round. Out-of-genus, none of the four clears chance and ProkBERT anti-ranks strongly (0.17), ranking the shuffle above the real promoter. The finding is unchanged in spirit, generalization holds in-genus and is lost past the genus, which is why *C. butyricum*'s in-genus status matters; the small engineered-library samples (68 and 25) keep it directional. Note this uses the harder held-out engineered positives, whereas the well-powered in-distribution motif test (Table 2) shows all four models clearing chance decisively.

5 Regime C: genome-scale precision (held-out in-genus)

Regimes A and B score curated test sets, in which positives are a large, fixed fraction of the windows. Neither shows the deployment reality, where real promoters are only about 0.01–0.07% of windows (Table 3), so at that rarity the question that decides whether a model is usable is its precision: of the sites it flags, what fraction are real. This regime slides each model across a whole held-out in-genus genome (*C. ljungdahlii*, Al-Bassam 2018, 1,245 primary TSS, never trained on) and, at each recall level, measures that precision (equivalently, how many false-positive peaks it produces per real promoter recovered), both on the full genome (9.26M windows) and restricted to the gene-upstream and intergenic candidate space (1.77M windows, 19.2% of the genome). Thresholds are chosen per model to hit each recall target, so the miscalibration of any single model (CNN-v2 in particular) does not distort the comparison.

Four findings follow from Table 3.

(1) Restricting the scan is the largest low-cost lever, and it helps every model. Scoring only the upstream and intergenic candidate space covers 19.2% of the genome and costs just 11 unreachable TSS, leaving a maximum recall of 99.1%, yet it raises precision across the board. The effect is largest for NT-LoRA, whose false-positive peaks at 50% recall fall by about 69% (from 4,139 to 1,294) because most of its errors lie outside the candidate space, in coding sequence. ProkBERT loses about 54% of its FP peaks (2,753 to 1,257), CNN-v2 about 41% (1,332 to 782), and the from-scratch CNN gains least, about 25% (1,284 to 967), because more of its FP peaks already lie inside the candidate space.

(2) In the restricted candidate space, the space that mining actually uses, CNN-v2 is the best model at every operationally relevant recall. At 50% recall it is the most precise (44.3%) with the fewest FP peaks (782), and it leads at 70% and 90% recall as well. The from-scratch CNN is second and is the most precise at the very-high-confidence 10% shortlist point (63.0% against CNN-v2's 59.8%). The two

Table 3: Genome-scale precision benchmark on *C. ljungdahlii* (both strands; full-scan base rate 0.013%, restricted 0.070%). Precision at four recall targets (0.1 is the high-confidence "shortlist" point) and false-positive peaks at 50% recall. **Bold** = best per column within each scan.

		Precision @ recall target				FP peaks
Model	Scan	0.1	0.5	0.7	0.9	@0.5
ProkBERT	full	23.6%	18.5%	15.8%	11.5%	2,753
NT-LoRA	full	18.4%	13.1%	10.8%	7.9%	4,139
CNN	full	**58.3%**	**32.7%**	**23.9%**	**12.7%**	**1,284**
CNN-v2	full	50.8%	31.9%	22.6%	11.5%	1,332
ProkBERT	restrict	38.6%	33.1%	29.6%	24.3%	1,257
NT-LoRA	restrict	51.7%	32.5%	27.9%	22.5%	1,294
CNN	restrict	**63.0%**	39.2%	33.7%	24.9%	967
CNN-v2	**restrict**	59.8%	**44.3%**	**36.8%**	**26.5%**	**782**

transformers trail: ProkBERT and NT are within a point of each other at mid-recall (33.1% and 32.5% at 50%), and ProkBERT is in fact the least precise of all four at the 10% shortlist (38.6%). The CNN family owns the restricted scan.

(3) Balanced AUPRC does not predict genome-scale precision, and this round shows it twice. NT-LoRA is second on balanced AUPRC (0.854) yet produces the most false positives of any model on the full genome (4,139 at 50% recall) and the lowest full-scan precision throughout. ProkBERT is first on balanced AUPRC (0.884) yet falls to third or fourth in the restricted scan and has the least precise high-confidence shortlist. Whatever lifts a model's curated-set score does not automatically survive the genome-scale base rate, which is the failure mode a balanced metric hides and the main reason for running this benchmark. Precision throughout is a lower bound, because the 1,245-primary-TSS reference omits the study's internal, antisense, and orphan TSS, so some peaks counted as false are genuine.

(4) The ProkBERT genome-scale numbers regressed relative to the earlier 0.9-clustered models. On the previous checkpoint ProkBERT's restricted precision at 50% recall was about 45% with roughly 760 FP peaks; on the 0.6 retrain it is 33.1% with 1,257 FP peaks, even though its validation AUPRC barely moved (0.876 to 0.878). Run through the identical candidate space, truth set, and scan pipeline, NT and the from-scratch CNN did not regress (their restricted numbers are stable or slightly better), so this is specific to the ProkBERT checkpoint rather than a pipeline artifact. Re-scanning the pre-0.6 checkpoint (T18) through the identical pipeline reproduced its 45.0% precision and 760 FP peaks at 50% recall, confirming the drop. Notably, that pre-0.6 ProkBERT was competitive with the CNN family at genome scale (45.0% at 50% recall, near CNN-v2's 44.3%), so the retrain, while it left the balanced numbers essentially unchanged, materially weakened ProkBERT for deployment. The Interpretation offers a hypothesis for the mechanism.

On this evidence the best standalone deployment configuration in this round is CNN-v2 with restrict-scan: it has the most precise mid-recall shortlist and the fewest false positives per true promoter across recall 0.5 to 0.9. The from-scratch CNN is a close second and marginally better at the 10% shortlist. The transformers are not the deployment pick at genome scale this round, which inverts the previous report. NT-LoRA is again the weakest member at genome scale and would not be used for standalone mining. The intended deliverable is still the four-model mean-rank-percentile ensemble; whether it beats CNN-v2 alone at genome scale has not yet been measured, and the two CNNs' correlation (below) is a caveat for how much the fourth vote adds.

6 Interpretation

CNN-v2 is the standout of this round. It is first at the composition-matched motif test (SRS AUROC 0.924, the cleanest signal that a model reads the motif rather than base composition), first in the restricted genome-scale scan at every practical recall, and it beats the from-scratch CNN on the balanced test set too (test AUPRC 0.848 against 0.804). A model that is only third on balanced AUPRC being first where it matters most, motif isolation and genome-scale precision, is the clearest example in this project of why the three regimes are reported separately. The added ENAC positional and k-mer features appear to buy real motif sensitivity rather than just fitting the training distribution.

CNN-v2 is miscalibrated, not weak. Its raw scores sit systematically higher than the other models', so at a naive 0.5 threshold it rejects fewer negatives (overall rejection 0.766, the lowest of the four) and recalls more positives. This is a calibration effect, not a discrimination one: its per-class AUROCs are among the best. It matters for two downstream steps. Calibration (T26) should be applied before any fixed-threshold use of its scores. The rank-percentile ensemble (T29), by contrast, is calibration-invariant, so CNN-v2's strong ranking enters the ensemble cleanly regardless of its score scale. The genome-scale benchmark already controls for this by choosing thresholds at fixed recall, which is why CNN-v2's miscalibration does not cost it there.

Genome scale is the deployment reality, and it reorders the models. Regimes A and B measure discrimination on curated test sets; Regime C measures what happens when a model runs across a 4.6 Mb genome in which promoters are about 0.01% of windows, and the two do not agree. NT-LoRA ranks second on balanced AUPRC but produces the most false positives and the lowest full-scan precision. ProkBERT ranks first on balanced AUPRC but is third or fourth in the restricted scan. The CNN family, lower on balanced AUPRC, is the most precise where promoters actually occur. The ranking that matters for deployment is precision in the gene-upstream candidate space, and there CNN-v2 is first. Two levers account for most of the achievable improvement, and neither is a change of model: restricting the search space is worth roughly 10 to 15 precision points at 50% recall in exchange for about 0.9% of recall, and moving to the 10% shortlist lifts precision further. Even so, the best configuration still returns a list that is roughly half false at moderate recall. This is what "enriched candidate list, not a map" means: a few-hundred-fold enrichment over blind guessing that a wet lab can test, not a finished annotation.

Why the 0.6 retrain may have hurt ProkBERT specifically. The stricter clustering removed more near-duplicate positives from training. One hypothesis is that a high-capacity pretrained language model such as ProkBERT leaned on those near-duplicates in a way that helped its genome-scale score distribution, whereas the low-capacity from-scratch CNN and the few-parameter NT LoRA adapter were less affected. This is a hypothesis about the mechanism, not a demonstrated one, and the balanced AUPRC being unchanged means the effect is invisible without the genome-scale benchmark. The effect itself is confirmed: re-scanning the pre-0.6 ProkBERT (T18) through the identical pipeline reproduced its 45% precision at 50% recall, against the 0.6 model's 33%. Because T18 also differs in its unswept hyperparameters, the regression is attributable to the retrain as a whole, not to the clustering threshold in isolation. This raises whether to deploy the stronger 0.9-clustered ProkBERT despite its weaker leakage guarantee, given that the genome-scale benchmark runs on a held-out genome and is not itself inflated by train/test near-duplicate leakage. CNN-v2 largely resolves the question: retrained at 0.9 its genome-scale precision is essentially unchanged (40% against 44% at 0.6, at 50% recall), so unlike ProkBERT it pays no genome-scale penalty for the stricter clustering, which is what one expects of a model reading genuine motif signal rather than memorising near-duplicates. Because CNN-v2 at 0.6 already matches the old 0.9 ProkBERT's genome-scale precision (44% against 45%) with none of the leakage exposure, the practical resolution is to lean on CNN-v2 at 0.6 rather than revert ProkBERT to 0.9.

What it means for the target. *C. butyricum* is in-genus, so the relevant discrimination evidence is the in-genus results, which are strong and motif-driven in Regimes A and B for all four models. Regime C sets expectations for the actual mine: score the target over the restricted candidate space, expect the CNN family (CNN-v2 in particular) to give the most enriched shortlist, and treat the output as a ranked list to validate, not a definitive map. The practical takeaway is to trust the in-genus numbers and let the genome-scale precision set how large a wet-lab shortlist to expect; out-of-genus behaviour is out of scope.

7 Caveats

- The ProkBERT genome-scale regression is confirmed by re-scanning the pre-0.6 checkpoint (T18) through the identical pipeline (45% against 33% precision at 50% recall). It still rests on one genome without CIs, and because T18 differs from the 0.6 model in both clustering threshold and hyperparameters, the regression is attributable to the retrain as a whole, not the 0.6 threshold in isolation.

- The four-model ensemble contains two CNNs (the from-scratch one-hot CNN and CNN-v2) that share an architecture family and were trained on the same data, so they are correlated. In a mean-rank-percentile ensemble this double-weights the CNN view and adds less diversity than a fourth independent model would. Whether to keep both, or let CNN-v2 replace the from-scratch CNN, should be decided by an explicit ensemble test at genome scale.
- Regime B's out-of-genus generalization numbers (*C. phytofermentans*) are now on the 0.6 models for all four, but rest on small engineered-library samples (68 in-genus, 25 out-of-genus) with wide CIs, so they are directional, not definitive. The in-distribution SRS test (Table 2) is the well-powered composition-matched result and should carry the weight of the motif-learning claim.
- Regime C rests on a single held-out genome and one study's TSS calls, graded with a 40 bp match tolerance and reported without CIs, and its precision is a lower bound because non-primary TSS are excluded. Because the excluded non-primary TSS may be distributed unevenly across models, the ranking itself, not only the absolute precision, could shift, and it should be reconfirmed on a second in-genus genome before being treated as settled.
- The regimes are not interchangeable: Regime A's heterogeneous negatives are the easiest, Regime B's composition-matched shuffle is the strictest balanced negative, and Regime C is the only one at the true genome base rate. Each should be read for the question it answers, not as a competing estimate of one number.